

FORGE · PRODUCTION TRACKING PLATFORM

Software Requirements Specification

Structured to ISO/IEC/IEEE 29148:2018. Defines the external interfaces, system features, functional and non-functional requirements, and verification approach for the Forge platform.

INTERNAL

PREPARED FOR

Symbiosys Technologies

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0.1 Document control

Revision history

VERSION	DATE	AUTHOR	SUMMARY OF CHANGE
0.1	Sep 2026	Engineering	Structure established against ISO/IEC/IEEE 29148:2018.
1.0	Sep 2026	Engineering	First complete specification: interfaces, 15 system features, non-functional requirements, data model, and verification matrix.

Approval

ROLE	NAME	SIGNATURE	DATE
Engineering Lead			
Quality Assurance			
Head of Product			

REQUIREMENT CONVENTIONS

Requirements use **shall** for mandatory behaviour, **should** for recommended behaviour, and **may** for optional behaviour, per the standard. Each requirement is atomic, uniquely identified, and verifiable by at least one method named in Section 7.

Priority is recorded as **HIGH**, **MEDIUM** or **LOW**. Every requirement traces to a business requirement in SYM-FRG-REQ-001; the matrix is in Appendix C.

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1 Introduction

1.1 Purpose

This document specifies the software requirements for Forge, a production tracking and review platform for animation and visual effects production. It is the engineering document of record: it defines what the software shall do in terms precise enough to design against, test against, and audit.

Its intended readers are the engineering team implementing and maintaining the system, quality assurance staff deriving test cases, and any party auditing the system's behaviour against its stated requirements.

1.2 Scope

Forge is a self-hosted web application that manages the production of visual content from initial tracking through departmental work, internal review and approval, to external client delivery. It replaces spreadsheet-based shot tracking and unstructured review feedback with a single system of record.

The software covers: identity and access management; the production hierarchy of projects, episodes, sequences, shots and assets; a tracking grid with spreadsheet import; task creation, assignment and progression; version management and media upload; a frame-accurate review player with annotation; a multi-stage approval chain; external client review by access link; attendance and effort recording; and studio administration with audit.

The software explicitly does not cover content creation application integration, render farm submission, media transcoding, financial ledger integration, or cross-tenant federation. These exclusions are recorded with rationale in SYM-FRG-REQ-001 Table 4.2.

1.3 Definitions, acronyms and abbreviations

TERM	DEFINITION AS USED IN THIS SPECIFICATION
Actor	A user or external system that initiates an interaction with the software.
Asset	A reusable production element tracked in parallel to shots.
Capability	A named permission, granted to a role, and evaluated per request to authorise an action.
Client session	An authenticated session established by an external reviewer through an access link, holding no user identity.
DCC	Digital Content Creation application.
Episode	An instalment of an episodic production, containing sequences.
JWT	JSON Web Token, the session token format.
ORM	Object-Relational Mapper; here, Prisma.
RBAC	Role-Based Access Control.
Scope	The subset of records a user may see, resolved as studio-wide, department, or own.
Sequence	A group of related shots within an episode.
Shot	The atomic unit of production work: a continuous run of frames.
Studio-wide role	A role whose visibility scope covers every record in the tenant.
Tenant	An isolated studio instance; every record belongs to exactly one.
Token version	An integer in both the session token and the user record; a mismatch invalidates the session.
Tracksheet	A spreadsheet used by a studio to track shot progress.
Version	A numbered iteration of deliverable media for a shot or asset.

Table 1.1 — Definitions.

1.4 References

REFERENCE	TITLE
ISO/IEC/IEEE 29148:2018	Systems and software engineering — Life cycle processes — Requirements engineering
SYM-FRG-REQ-001	Forge — Requirements Document
SYM-FRG-USG-001	Forge — Product Usage & User Guide
SYM-FRG-MKT-001	Forge — Marketing Analysis
RFC 9106	Argon2 Memory-Hard Function for Password Hashing and Proof-of-Work Applications
RFC 7519	JSON Web Token

Table 1.2 — Referenced documents and standards.

1.5 Document overview

Section 2 describes the product in context. Section 3 specifies the external interfaces. Section 4 specifies system features and their functional requirements, and is the largest section. Section 5 specifies non-functional requirements. Section 6 specifies data requirements. Section 7 defines how each requirement class is verified. Appendices provide the capability reference, configuration parameters, and the traceability matrix.

2 Overall description

2.1 Product perspective

Forge is a self-contained system, not a component of a larger product. It replaces a manual process rather than an existing system. It comprises a browser-delivered client application, an HTTP application programming interface, a relational database, an in-memory cache, and a reverse proxy, deployed together as containers on a single host controlled by the studio.

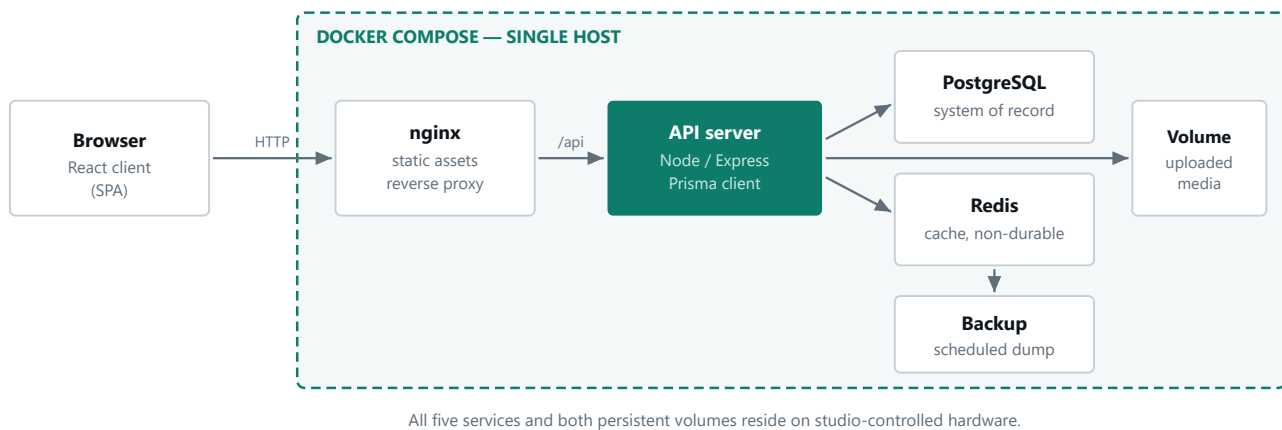


Figure 2.1 — Deployment architecture. The API tier is stateless; all durable state is in PostgreSQL and the media volume.

2.2 Product functions

At summary level the software shall provide: authentication and account lifecycle management; capability-based authorisation with scoped visibility; management of the production hierarchy; a filterable tracking grid with spreadsheet import and export; task lifecycle management; version and media management; frame-accurate review with annotation; a multi-stage approval chain; external client review; attendance and effort recording; notification and messaging; and administration with an auditable change history.

2.3 User classes and characteristics

CLASS	FREQUENCY OF USE	TECHNICAL EXPERTISE	PRIVILEGE SUMMARY
Artist	Daily, sustained	Domain expert, non-technical	Own work only; edit and submit; no assignment or approval authority.
Lead	Daily	Domain expert, some technical	Own department; create, assign, approve at lead stage.
Production Head	Daily	Domain expert	Studio-wide; full production authority plus member administration.
Producer	Daily	Domain expert	Studio-wide; final approval gate; administrative capability.
Administrator	Occasional	Technical	Accounts, roles, licences, integrations, pipeline configuration; no task or review capability.
External client	Intermittent	Non-technical	Only items explicitly shared; comment and review decision.

Table 2.1 — User classes.

2.4 Operating environment

ELEMENT	REQUIREMENT
Server operating system	Any operating system supporting Docker Engine and Docker Compose.
Container runtime	Docker Engine with Compose v2.
Database	PostgreSQL 15.
Cache	Redis 7, persistence disabled.
Application runtime	Node.js 24 within the API container.
Client	Evergreen desktop browser with HTML5 video and Canvas support.
Network	Studio local network. Outbound internet not required for core function.

Table 2.2 — Operating environment.

2.5 Design and implementation constraints

- **CON-1.** The system shall be deployable entirely on studio-controlled hardware.
- **CON-2.** Core function shall not depend on outbound internet access, prohibiting runtime reliance on external content delivery networks or third-party APIs.
- **CON-3.** All studio traffic may present a single network source address; protective controls shall not treat the studio as one abusive client.
- **CON-4.** The system holds live production data; schema changes shall be verified against a clone restored from production before deployment.
- **CON-5.** Verification of any change shall exercise the same network path as real traffic.
- **CON-6.** The implementation language shall be TypeScript across all tiers.
- **CON-7.** Database access shall be through the Prisma client; raw SQL is permitted only where the client cannot express the query, and shall be parameterised.

2.6 User documentation

The software shall be accompanied by SYM-FRG-USG-001, structured so that each role has a self-contained access path covering only the functionality available to that role, plus installation, configuration and troubleshooting material for the deploying administrator.

2.7 Assumptions and dependencies

Assumptions ASM-1 to ASM-6 and dependencies are specified in SYM-FRG-REQ-001 Sections 11.2 and 13, and are incorporated here by reference. The dependency of note for this specification is the SMTP relay: its unavailability shall degrade password recovery and client access-code delivery without affecting any other function.

3 External interface requirements

3.1 User interfaces

ID	REQUIREMENT	PRI.
UI-001	The client application shall be delivered as a single-page application over HTTP and shall require no installation on the user's workstation.	HIGH
UI-002	On successful authentication the system shall route each user to a landing view determined by their role.	HIGH
UI-003	Navigation shall present only destinations the authenticated user's role is permitted to reach.	HIGH
UI-004	Controls invoking an action the user lacks capability for shall not be rendered.	HIGH
UI-005	Where the absence of a control would be ambiguous, the interface shall state the reason the action is unavailable.	MED
UI-006	An empty result set shall indicate where related data may be found rather than presenting a terminal state.	MED
UI-007	The interface shall render correctly in both light and dark presentation.	LOW
UI-008	All interactive controls shall be reachable by keyboard and shall present a visible focus indicator.	MED
UI-009	The review player shall provide keyboard shortcuts for transport and tool selection.	MED

Table 3.1 — User interface requirements.

3.2 Hardware interfaces

The software has no direct hardware interface. It requires host storage for the database volume and the media volume, sized in Appendix B. No specialised hardware, dongle or licence device is required.

3.3 Software interfaces

ID	REQUIREMENT	PRI.
SI-001	The API shall persist all durable state to PostgreSQL 15 through the Prisma client.	HIGH
SI-002	The API shall use Redis for authorisation caching, rate-limit counters and presence throttling, and shall treat all Redis contents as reconstructible.	HIGH
SI-003	The API shall deliver electronic mail through an SMTP relay configured by environment variable.	HIGH
SI-004	Schema migrations shall be applied by the migration runner at API container start, and shall be idempotent.	HIGH
SI-005	The system may export error and performance telemetry to an external monitoring service; its absence shall not affect function.	LOW

Table 3.2 — Software interface requirements.

3.4 Communication interfaces

ID	REQUIREMENT	PRI.
CI-001	Client-to-server communication shall use HTTP with JSON request and response bodies, except file upload which shall use multipart form encoding.	HIGH
CI-002	The session token shall be transmitted in an HTTP cookie marked HttpOnly and SameSite=Lax.	HIGH
CI-003	The session cookie shall be marked Secure when the deployment is configured for transport encryption.	HIGH
CI-004	The API shall be reachable only through the reverse proxy, which shall forward the originating client address.	HIGH
CI-005	The API shall trust exactly one proxy hop when resolving the client address, so that a client cannot forge its apparent origin.	HIGH
CI-006	Media responses shall support HTTP range requests to permit seeking without full download.	HIGH

Table 3.3 — Communication interface requirements.

NOTE ON CI-005

The number of trusted proxy hops is security-relevant. Trusting more hops than exist allows a client to inject a forged forwarded-address header and defeat address-based rate limiting; trusting fewer collapses every client onto the proxy's own address. The deployment has exactly one proxy, and the configuration shall state one.

4 System features

Fifteen features, each with a description, a priority, its stimulus and response behaviour, and its functional requirements. Requirement identifiers correspond to those in SYM-FRG-REQ-001 Section 8 where a direct counterpart exists.

4.1 Authentication

PRIORITY High	TRACES TO BR-012	ACTORS All internal users
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Description. Establishes and verifies user identity, and manages the credential lifecycle including self-registration and recovery.

Stimulus and response. A user submits an email address and password; the system verifies the credential and, on success, establishes a session and returns the user's identity, role and capability set. On failure the system returns an indistinguishable error regardless of whether the account exists.

ID	REQUIREMENT	PRI.
SRS-AUTH-001	The system shall authenticate a user by email address and password.	HIGH
SRS-AUTH-002	The system shall store passwords only as Argon2id hashes and shall never store or log a plaintext password.	HIGH
SRS-AUTH-003	On authentication failure the system shall return a message that does not distinguish an unknown account from an incorrect password.	HIGH
SRS-AUTH-004	When the submitted email address matches no account, the system shall perform a hash verification against a decoy value so that response time does not disclose account existence.	HIGH
SRS-AUTH-005	The system shall refuse authentication to an account whose status is not active, and shall state that the account is deactivated.	HIGH
SRS-AUTH-006	The system shall permit an unauthenticated visitor to register an account, collecting name, email address, password, department and a requested role.	HIGH
SRS-AUTH-006a	The system shall provide an unauthenticated endpoint listing the departments and role names available at registration, rate limited on the registration budget.	MED
SRS-AUTH-006b	A self-selected department shall be validated as belonging to the registering tenant before it is stored.	HIGH
SRS-AUTH-006c	The system shall create every self-registered account at the least-privilege role, irrespective of the role requested.	HIGH
SRS-AUTH-006d	Where the requested role carries more authority than the least-privilege role, the system shall record the request against the account for an administrator to decide.	HIGH
SRS-AUTH-006e	An unrecognised requested role shall be discarded without failing the registration, so the field cannot be used to probe which roles exist.	MED
SRS-AUTH-006f	A pending role request shall be visible only to actors whose visibility scope is studio-wide.	HIGH
SRS-AUTH-006g	An administrator setting a user's role shall clear any pending role request on that account.	HIGH
SRS-AUTH-007	The system shall respond to a registration attempt for an already-registered address with the same message as a successful registration.	HIGH
SRS-AUTH-008	The system shall permit a user to request a password reset by email address.	HIGH
SRS-AUTH-009	The system shall store password reset tokens only as SHA-256 hashes.	HIGH
SRS-AUTH-010	A password reset token shall be single-use and shall expire 30 minutes after issue.	HIGH
SRS-AUTH-011	Consuming a password reset token shall invalidate all existing sessions for that user.	HIGH
SRS-AUTH-012	The system shall permit an authenticated user to change their own password on presentation of the current password.	HIGH
SRS-AUTH-013	A successful password change shall invalidate all other sessions for that user.	HIGH

Table 4.1 — Authentication requirements.

RATIONALE FOR SRS-AUTH-006C

Registration is reachable by anyone who can load the sign-in page. If the role field were honoured as submitted, that person could grant themselves studio-wide authority by choosing it from a dropdown — the most direct privilege escalation the system could offer. Separating the request from the grant keeps onboarding self-service without making authority self-service: the new starter can sign in and work immediately, and the only thing that waits on an administrator is the elevation itself.

4.2 Session management

PRIORITY High	TRACES TO BR-007	ACTORS All users
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Description. Maintains authenticated state across requests and provides server-side revocation of a stateless token.

ID	REQUIREMENT	PRI.
SRS-SESS-001	The system shall represent a session as a signed JWT conveyed in an HttpOnly cookie.	HIGH
SRS-SESS-002	A session token shall expire 7 days after issue.	HIGH
SRS-SESS-003	The session token shall carry the tenant identifier, user identifier, role identifier, department identifier and token version.	HIGH
SRS-SESS-004	The system shall reject any request whose token version does not equal the current token version of the referenced user.	HIGH
SRS-SESS-005	The system shall reject any request whose referenced user is not active.	HIGH
SRS-SESS-006	The system shall increment a user's token version on logout, password change, password reset, role change and deactivation.	HIGH
SRS-SESS-007	Session validity data may be cached for no longer than 300 seconds, and the cache entry shall be deleted when the token version is incremented.	HIGH
SRS-SESS-008	If session validity cannot be determined, the system shall reject the request.	HIGH
SRS-SESS-009	An active session shall survive loss of the cache without requiring re-authentication.	HIGH

Table 4.2 — Session management requirements. SRS-SESS-008 and SRS-SESS-009 together specify fail-closed validity with fail-soft performance.

DESIGN NOTE

SRS-SESS-004 is what makes a stateless token revocable. Without it, logout would clear only the browser's copy of a token that remains valid for its full seven-day life — a token captured from a shared workstation would continue to work after the user believed they had signed out.

4.3 Access control

PRIORITY	TRACES TO	ACTORS
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High	BR-007, BR-011	All users
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Description. Authorises each action against a named capability granted to the actor's role.

ID	REQUIREMENT	PRI.
SRS-RBAC-001	The system shall evaluate authorisation against a named capability held by the actor's role, retrieved from persistent storage.	HIGH
SRS-RBAC-002	The system shall not authorise any action on the basis of a hard-coded role name comparison.	HIGH
SRS-RBAC-003	The system shall enforce authorisation on the server for every request, independently of any client-side concealment.	HIGH
SRS-RBAC-004	The system shall return HTTP 403 with no resource detail when an authenticated actor lacks the required capability.	HIGH
SRS-RBAC-005	The administrator role shall hold none of: create_tasks, edit_tasks, assign_tasks, delete_tasks, submit_reviews, approve_reviews.	HIGH
SRS-RBAC-006	The system shall permit an authorised administrator to modify capability grants at runtime without redeployment.	MED
SRS-RBAC-007	A capability change shall take effect on the affected users' next request.	HIGH
SRS-RBAC-008	The system shall refuse navigation to an application route not permitted to the actor's role.	HIGH

Table 4.3 — Access control requirements. The complete capability reference is Appendix A.

4.4 Multi-tenancy and isolation

PRIORITY	TRACES TO	ACTORS
High	BR-007	System

ID	REQUIREMENT	PRI.
SRS-TEN-001	Every production record shall carry a tenant identifier.	HIGH
SRS-TEN-002	Every query shall be constrained by the tenant identifier of the requesting session.	HIGH
SRS-TEN-003	A request for a record belonging to another tenant shall be answered as not found, not as forbidden.	HIGH
SRS-TEN-004	File retrieval paths shall include the tenant identifier and shall refuse a mismatch.	HIGH

Table 4.4 — Tenancy requirements. SRS-TEN-003 prevents cross-tenant existence disclosure.

4.5 Visibility scoping

PRIORITY	TRACES TO	ACTORS
High	BR-007	All internal users

Description. Determines, within a tenant, which records an actor may see. Distinct from capability: capability governs what kind of action is permitted, scope governs which records it may be applied to.

ID	REQUIREMENT	PRI.
SRS-SCP-001	The system shall resolve an actor's visibility scope as one of: all, department, or own.	HIGH
SRS-SCP-002	Roles designated studio-wide shall resolve to scope "all".	HIGH
SRS-SCP-003	A lead shall resolve to scope "department", limited to their assigned department.	HIGH
SRS-SCP-004	An artist shall resolve to scope "own".	HIGH
SRS-SCP-005	Holding a capability shall not extend an actor's scope; an actor may only act on records within scope.	HIGH
SRS-SCP-006	Personal data attached to a user record, including contact address and attendance times, shall be returned only to actors with a supervisory relationship or to the subject.	HIGH
SRS-SCP-007	A client session shall resolve to a scope containing only records explicitly shared with its access link.	HIGH

Table 4.5 — Visibility scoping requirements.

NOTE ON SRS-SCP-005

Capability and scope must both be checked. Holding edit_tasks states what kind of change an actor may make, not which tasks they may make it to; without an independent scope check an artist holding edit_tasks could modify any task in the tenant, including work in departments they cannot list.

4.6 Production structure

PRIORITY High	TRACES TO BR-001	ACTORS Production management
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ID	REQUIREMENT	PRI.
SRS-STR-001	The system shall represent projects, and within a project, episodes, sequences and shots in a strict hierarchy.	HIGH
SRS-STR-002	The system shall represent assets as first-class records parallel to shots.	HIGH
SRS-STR-003	The system shall represent departments with an ordered pipeline position.	HIGH
SRS-STR-004	Deleting a parent record shall cascade to its children or be refused; orphaned children shall not result.	HIGH
SRS-STR-005	Shot identifiers shall be unique within a project.	HIGH

Table 4.6 — Production structure requirements.

4.7 Tracking grid

PRIORITY High	TRACES TO BR-001	ACTORS Leads, production management
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ID	REQUIREMENT	PRI.
SRS-GRD-001	The system shall present a tabular view of shots showing project, episode, sequence, shot, assignee, status, version, and review states.	HIGH
SRS-GRD-002	The grid shall be filterable by project, episode, sequence, artist, position and department.	HIGH
SRS-GRD-003	The grid shall permit in-place editing of status and review state by actors holding edit capability.	HIGH
SRS-GRD-004	The grid shall present only records within the actor's visibility scope.	HIGH
SRS-GRD-005	The system shall permit a grid configuration to be saved and recalled as a named view.	MED
SRS-GRD-006	The system shall export the current grid view as CSV.	MED
SRS-GRD-007	Pending in-place edits shall be committed explicitly and shall be discardable before commit.	MED

Table 4.7 — Tracking grid requirements.

4.8 Spreadsheet import

PRIORITY High	TRACES TO BR-004	ACTORS Production management
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Description. Ingests an existing studio tracksheet without requiring the studio to reformat it. Its governing principle is tolerance: accept what is given, and never discard what is not understood.

Stimulus and response. An authorised user selects a project and supplies a workbook. The system parses each sheet, creates or reuses episodes, sequences and shots, creates a task per row, and reports the outcome of every row.

ID	REQUIREMENT	PRI.
SRS-IMP-001	The system shall accept workbooks in .xlsx, .xls and .csv formats.	HIGH
SRS-IMP-002	Where a sheet name matches the pattern for an episode designation, the system shall derive the episode from the sheet name.	HIGH
SRS-IMP-003	Where no sheet name matches that pattern, the system shall import any sheet containing recognisable shot rows under a default episode.	HIGH
SRS-IMP-004	The system shall skip a sheet containing no recognisable shot column, and shall not create records from it.	HIGH
SRS-IMP-005	Column matching shall be insensitive to case, whitespace, underscores and hyphens, and shall accept a defined set of synonyms per field.	HIGH
SRS-IMP-006	The system shall preserve every non-empty column it does not map to a known field, recorded against the created task.	HIGH
SRS-IMP-007	The system shall limit the parsed row range to a bounded maximum irrespective of the range the sheet declares.	HIGH
SRS-IMP-008	The system shall parse both slash-separated and eight-digit day-month-year date forms, and shall record no date rather than an incorrect one where a value cannot be confidently parsed.	HIGH
SRS-IMP-009	The system shall attempt to match a named artist to an existing user by whole-word comparison of all name parts.	MED
SRS-IMP-010	The system shall report the outcome of each row as created, updated or skipped, with a reason for each skip.	HIGH
SRS-IMP-011	Where no shot row is found anywhere in the workbook, the system shall report this as a failure and shall state what was expected.	HIGH
SRS-IMP-012	Import shall be idempotent with respect to shots: re-importing a workbook shall reuse existing shots rather than duplicating them.	HIGH

Table 4.8 — Import requirements.

RATIONALE FOR SRS-IMP-007

A real production tracksheet was observed declaring a range extending to the spreadsheet format's maximum row count, because a single stray value existed at that row. Parsing the declared range materialised approximately one million mostly-empty row objects and took over thirty seconds, which presents as a hung application. The bound is a correctness requirement, not an optimisation.

RATIONALE FOR SRS-IMP-011

Reporting "0 of 0 rows imported" as a success is indistinguishable from a silent failure to the user. The system shall state that no shot column was recognised and name the headers it looks for.

4.9 Task management

PRIORITY High	TRACES TO BR-001, BR-011	ACTORS All internal users
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ID	REQUIREMENT	PRI.
SRS-TSK-001	An actor holding create_tasks shall create a task against a shot or asset.	HIGH
SRS-TSK-002	Changing a task's assignee shall require assign_tasks; changing any other field shall require edit_tasks.	HIGH
SRS-TSK-003	An actor may assign an unassigned task to themselves without holding assign_tasks; this exemption shall apply only where the task is currently unassigned and the assignee set is the actor.	MED
SRS-TSK-004	A task modification shall be refused where the task lies outside the actor's visibility scope, irrespective of capability.	HIGH
SRS-TSK-005	The system shall validate that a nominated assignee belongs to the same tenant.	HIGH
SRS-TSK-006	The system shall permit any actor who can view a task to post a comment on it, without requiring edit_tasks.	HIGH
SRS-TSK-007	A client session shall not post task comments.	HIGH
SRS-TSK-008	The system shall support checklist items, file attachments and inter-task dependencies.	MED
SRS-TSK-009	The system shall present tasks in both list and board arrangements.	MED
SRS-TSK-010	On approval, the system shall offer handoff of the task to the next department in pipeline order.	MED

Table 4.9 — Task management requirements.

RATIONALE FOR SRS-TSK-006

Commenting is an oversight action, not production work. Requiring edit_tasks to comment would prevent the administrator role — which by SRS-RBAC-005 holds no task capability — from asking a question about work it is responsible for monitoring. Visibility is the correct gate; capability is not.

4.10 Versioning and media

PRIORITY High	TRACES TO BR-002, BR-005	ACTORS Artists, leads
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ID	REQUIREMENT	PRI.
SRS-VER-001	The system shall record versions against a shot or asset, each with an ordinal and a review status.	HIGH
SRS-VER-002	Media upload shall require submit_reviews.	HIGH
SRS-VER-003	The system shall accept video uploads of MP4, WebM, QuickTime and AVI types, determined by declared media type, and shall reject all others.	HIGH
SRS-VER-004	Video uploads shall be limited to 500 MB; general file attachments shall be limited to 200 MB.	HIGH
SRS-VER-005	Uploaded files shall be stored under a generated identifier, never under the name supplied by the client.	HIGH
SRS-VER-006	Uploaded media shall be written to a storage volume that persists independently of container lifecycle.	HIGH
SRS-VER-007	Video shall be served with its true media type and shall support range requests.	HIGH
SRS-VER-008	Non-video attachments shall be served with a generic binary media type, with content sniffing disabled and a disposition forcing download.	HIGH
SRS-VER-009	Uploaded media shall be associated with the version record so that it is visible to every actor who can view that version, not only to the uploader.	HIGH

Table 4.10 — Versioning and media requirements.

RATIONALE FOR SRS-VER-008

An uploaded HTML or SVG document served inline with its natural media type would execute as a same-origin page with access to the session cookie. Because attachments share an origin with the application, the client-declared type is never trusted to determine how a response is rendered. Video is the deliberate exception under SRS-VER-003, where the accepted type set is restricted at write time.

4.11 Review player

PRIORITY High	TRACES TO BR-002	ACTORS Artists, leads, production management, clients
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Description. The frame-accurate playback and annotation surface where review actually happens.

ID	REQUIREMENT	PRI.
SRS-PLR-001	The player shall provide play, pause, step-forward and step-backward transport, and shall display the current frame and total frame count.	HIGH
SRS-PLR-002	The player shall permit seeking to an arbitrary frame.	HIGH
SRS-PLR-003	The player shall provide freehand, arrow, rectangle and text annotation tools.	HIGH
SRS-PLR-004	The player shall provide a selection tool and an eraser that removes only the acting user's own annotations.	HIGH
SRS-PLR-005	The player shall provide a colour selection of at least six values applied to subsequently created annotations.	MED
SRS-PLR-006	Annotations shall be stored against a specific version and frame and shall persist across sessions.	HIGH
SRS-PLR-007	The player shall provide an A/B wipe comparing two versions across a draggable divider.	HIGH
SRS-PLR-008	The player shall provide onion skinning displaying adjacent frames semi-transparently.	HIGH
SRS-PLR-009	The player shall provide ghosting and side-by-side version comparison.	MED
SRS-PLR-010	The player shall capture the current frame including annotations to the system clipboard.	LOW
SRS-PLR-011	Comments posted from the player shall record the frame at which they were made and shall permit navigation back to that frame.	HIGH
SRS-PLR-012	The player shall provide a read-only mode in which no annotation or modification is possible.	MED
SRS-PLR-013	The system shall provide a presentation mode in which one presenter's frame position is followed by other viewers of the same version.	LOW
SRS-PLR-014	A viewer in a presenter-led session shall be prevented from altering playback position or annotations.	MED
SRS-PLR-015	Where no media has been uploaded for a version, the player shall state this and, for actors holding submit_reviews, offer upload.	MED
SRS-PLR-016	An actor lacking submit_reviews shall be shown the reason they cannot post feedback rather than a control that fails on use.	MED

Table 4.11 — Review player requirements.

4.12 Approval chain

PRIORITY High	TRACES TO BR-003	ACTORS Artists, leads, production management
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Description. The staged progression of work from artist submission to final approval, with a permanent record of each decision. The state model is specified in SYM-FRG-REQ-001 Figure 7.3.

ID	REQUIREMENT	PRI.
SRS-APR-001	The system shall support the stages: not started, in progress, lead review, production review, approved.	HIGH
SRS-APR-002	Advancing a task to production review shall require the actor to hold departmental leadership authority for that task's department.	HIGH
SRS-APR-003	Advancing a task to approved shall require the actor to hold production management authority.	HIGH
SRS-APR-004	Stage-advancing authority shall be enforced on the server; the interface shall not be the only gate.	HIGH
SRS-APR-005	Every stage transition shall create an immutable approval event recording actor, action, timestamp and subject version.	HIGH
SRS-APR-006	A rejection or change request shall return the task to in progress and shall record the reason.	HIGH
SRS-APR-007	The system shall present a review queue distinguishing items awaiting the actor from items at other stages.	HIGH
SRS-APR-008	Where no item awaits the actor, the queue shall indicate the number of items at other stages and provide navigation to them.	MED
SRS-APR-009	Approval history shall be displayed alongside the version it concerns.	HIGH

Table 4.12 — Approval chain requirements.

4.13 External client access

PRIORITY High	TRACES TO BR-008	ACTORS Production management, external clients
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ID	REQUIREMENT	PRI.
SRS-CLI-001	An authorised actor shall generate an access link granting an external party review access to nominated items.	HIGH
SRS-CLI-002	Redemption of an access link shall additionally require a code issued with it.	HIGH
SRS-CLI-003	The system shall deliver the access code to the nominated recipient by email on creation.	HIGH
SRS-CLI-004	Content in outbound mail derived from user input shall be escaped so that it cannot inject markup or additional mail headers.	HIGH
SRS-CLI-005	A client session shall carry no user identity and shall be distinguishable from an employee session by the system.	HIGH
SRS-CLI-006	A client session shall be refused access to every endpoint except those explicitly permitted to client access.	HIGH
SRS-CLI-007	A client session shall see only the items associated with its access link.	HIGH
SRS-CLI-008	Client feedback shall be recorded against the subject item and shall be visible to studio users who can view that item.	HIGH
SRS-CLI-009	An access link shall be revocable, and revocation shall take effect on the next request made with it.	HIGH
SRS-CLI-010	A client review decision shall be written through an endpoint scoped to client sessions, not through the general update path.	HIGH

Table 4.13 — Client access requirements.

4.14 Attendance and effort

PRIORITY	TRACES TO	ACTORS
Medium	BR-009	All internal users

ID	REQUIREMENT	PRI.
SRS-ATT-001	On successful authentication of a user in a production role, the system shall record a clock-in time if none is open.	HIGH
SRS-ATT-002	On sign-out, the system shall close any open clock-in for that user.	HIGH
SRS-ATT-003	Clock-in on authentication shall be idempotent; repeated sign-in shall not create overlapping records.	HIGH
SRS-ATT-004	The system shall record a last-seen timestamp for each authenticated user, updated on request activity.	MED
SRS-ATT-005	Last-seen updates shall be throttled to at most one database write per user per 45 seconds.	HIGH
SRS-ATT-006	A user shall be presented as online where their last-seen timestamp is within 90 seconds of the present.	MED
SRS-ATT-007	Presence shall not be derived from clock-in state alone.	HIGH
SRS-ATT-008	Presence and attendance data shall be returned only within the requesting actor's attendance visibility scope.	HIGH
SRS-ATT-009	Where an actor may not see another user's presence, the system shall present a neutral state rather than an inferred offline state.	MED
SRS-ATT-010	The system shall record time logged against individual tasks.	HIGH
SRS-ATT-011	The system shall support a daily standup record per user per day.	MED

Table 4.14 — Attendance and effort requirements.

RATIONALE FOR SRS-ATT-007

Clock-in is an attendance fact that survives closing a laptop; presence is a statement about current activity. Deriving one from the other means a user who closes their browser without signing out is displayed as online indefinitely. The heartbeat in SRS-ATT-004 to 006 is what separates them.

4.15 Administration and audit

PRIORITY High	TRACES TO BR-011, BR-013	ACTORS Administrators, production management
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ID	REQUIREMENT	PRI.
SRS-ADM-001	An actor holding manage_members shall create, deactivate and reactivate accounts.	HIGH
SRS-ADM-002	Deactivation shall preserve all records authored by or referring to the deactivated user.	HIGH
SRS-ADM-003	Deactivating an account or changing its role shall invalidate that user's existing sessions.	HIGH
SRS-ADM-004	An actor holding manage_roles shall modify capability grants.	HIGH
SRS-ADM-005	The system shall write an audit record for each significant state change, capturing actor, entity, action and timestamp.	HIGH
SRS-ADM-006	The system shall permit reversal of an audited change where reversal is safe, and shall record the reversal as its own audit event.	MED
SRS-ADM-007	Studio-level configuration shall be modifiable without redeployment.	MED
SRS-ADM-008	Entity schemas and task templates shall be configurable by authorised actors.	LOW
SRS-ADM-009	Features not yet implemented shall be visible only to administrators and shall be labelled as forthcoming rather than presented as operational.	MED

Table 4.15 — Administration and audit requirements.

5 Non-functional requirements

5.1 Performance requirements

ID	REQUIREMENT	PRI.
SRS-PRF-001	The system shall serve at least 100 concurrent authenticated sessions on the reference hardware without degradation of interactive response.	HIGH
SRS-PRF-002	The tracking grid shall render and remain interactive with at least 5,000 shot rows in scope.	HIGH
SRS-PRF-003	Session validity determination shall not require a database query on every request under normal cache operation.	HIGH
SRS-PRF-004	Presence tracking shall not generate a database write per request.	HIGH
SRS-PRF-005	Spreadsheet parsing shall complete within 10 seconds for a workbook of up to 5,000 data rows.	MED
SRS-PRF-006	Media playback shall begin without requiring the complete file to transfer.	HIGH

Table 5.1 — Performance requirements.

5.2 Security requirements

ID	REQUIREMENT	PRI.
SRS-SEC-001	Passwords shall be hashed with Argon2id.	HIGH
SRS-SEC-002	The system shall never write a plaintext password or session token to any log.	HIGH
SRS-SEC-003	All database access shall be parameterised; query text shall never be assembled by concatenating request data.	HIGH
SRS-SEC-004	Authentication attempts shall be rate limited per source address and per account independently.	HIGH
SRS-SEC-005	Only failed authentication attempts shall consume rate-limit budget.	HIGH
SRS-SEC-006	Registration, password reset request, reset consumption and password change shall each be independently rate limited.	HIGH
SRS-SEC-007	Rate limits shall be set so that a studio sharing one network egress address is not locked out by legitimate use.	HIGH
SRS-SEC-008	Where the rate-limiting store is unavailable, requests shall be permitted rather than denied.	MED
SRS-SEC-009	Session validity checks shall deny the request where validity cannot be determined.	HIGH
SRS-SEC-010	Uploaded file names supplied by the client shall never determine the storage path.	HIGH
SRS-SEC-011	File retrieval shall reduce any supplied name to its base component, preventing directory traversal.	HIGH
SRS-SEC-012	Outbound mail shall escape user-supplied content against markup and header injection.	HIGH
SRS-SEC-013	Error responses shall not disclose stack traces, query text or internal identifiers to the client.	HIGH
SRS-SEC-014	The system shall not disclose whether an email address corresponds to an account through any unauthenticated endpoint.	HIGH

Table 5.2 — Security requirements. Configured rate-limit values are in Appendix B.

RATIONALE FOR THE ASYMMETRY BETWEEN SRS-SEC-008 AND SRS-SEC-009

These two requirements deliberately fail in opposite directions. Rate limiting is an availability control: if its store is unavailable, denying every request would convert a cache outage into a total outage, so it fails open. Session validity is a security control: if it cannot be determined, permitting the request would allow a revoked session to continue, so it fails closed. Stating both explicitly prevents a future maintainer from "correcting" one to match the other.

5.3 Reliability and availability

ID	REQUIREMENT	PRI.
SRS-REL-001	Core function shall not require outbound internet access.	HIGH
SRS-REL-002	Containers shall restart automatically unless deliberately stopped.	HIGH
SRS-REL-003	The API shall not accept traffic until the database and cache report healthy.	HIGH
SRS-REL-004	Schema migrations shall run automatically at API start and shall be idempotent.	HIGH
SRS-REL-005	The database shall be dumped on a schedule to a volume independent of the database volume.	HIGH
SRS-REL-006	Uploaded media shall persist across container rebuild and image replacement.	HIGH
SRS-REL-007	Loss of the cache shall degrade performance only, and shall not invalidate sessions or alter authorisation outcomes.	HIGH
SRS-REL-008	A failure in a non-essential side effect, such as notification dispatch, shall not fail the primary operation.	MED

Table 5.3 — Reliability requirements.

5.4 Software quality attributes

ID	REQUIREMENT	PRI.
SRS-QUA-001	All application source shall be statically typed and shall pass type checking with no suppressed errors in the release build.	HIGH
SRS-QUA-002	Application logs shall be emitted as structured records.	MED
SRS-QUA-003	The request and response contract shall be defined once and shared between client and server rather than declared independently.	HIGH
SRS-QUA-004	Build and type-check tasks shall be independently cacheable per workspace package.	MED
SRS-QUA-005	Code implementing a non-obvious constraint shall record the reason for that constraint at the point of implementation.	MED

Table 5.4 — Quality attributes.

5.5 Business rules

ID	RULE
BRU-001	A task may advance to production review only through a departmental lead of that task's department.
BRU-002	A task may be approved only by production management.
BRU-003	The administrator role performs no production work and holds no task or review capability.
BRU-004	An artist sees only work assigned to them; a lead sees only their department; production management sees the studio.
BRU-005	An external client sees only what has been explicitly shared and never enters the studio application.
BRU-006	Production media remains on studio-controlled storage at all times.
BRU-007	Approval history is append-only and is never edited or deleted.
BRU-008	Deactivating a person removes their access but never their contribution history.

Table 5.5 — Business rules.

6 Data requirements

6.1 Logical data model

The system comprises 60 entity types. The production workflow is carried by the hierarchy specified in SYM-FRG-REQ-001 Figure 7.4; the remainder support communication, configuration, administration and audit.

GROUP	PRINCIPAL ENTITIES
Tenancy and identity	Tenant, User, TenantRole, TenantRoleCapability, PendingInvite, PasswordResetToken, PlatformUser
Production hierarchy	Project, Episode, Sequence, Shot, Asset, Department, SequenceTeamMember
Work	Task, TaskComment, TaskChecklistItem, TaskAttachment, TaskDependency, TaskApprovalEvent, TaskTemplate, TaskTemplateItem
Review	Version, Review, ReviewComment, ReviewPresentation, Annotation, ClientNote
Client delivery	ClientAccessLink, Delivery, DeliveryItem, DeliveryDownload, PublishLog
Time and activity	DailyLog, StandupUpdate, StandupApproval, StandupPlaylistItem, AssetActivity
Communication	Notification, NotificationPreference, ChatChannel, ChatChannelMember, ChatMessage, Broadcast
Pipeline configuration	PipelineTemplate, PipelineStage, ProjectPipeline, Workflow, WorkflowRun, EntityTypeDef, EntityFieldDef, TrackingView
Platform administration	StudioSetting, ApiKey, Webhook, Integration, TenantPlugin, LicenseServer
Audit	AuditLog, AuditRollback

Table 6.1 — Entity groups.

6.2 Data integrity requirements

ID	REQUIREMENT	PRI.
SRS-DAT-001	Referential integrity shall be enforced by the database through foreign key constraints, not by application logic alone.	HIGH
SRS-DAT-002	Every tenant-owned entity shall carry a non-null tenant identifier.	HIGH
SRS-DAT-003	User email addresses shall be unique across the system.	HIGH
SRS-DAT-004	Password reset token hashes shall be unique.	HIGH
SRS-DAT-005	Records supporting soft deletion shall be excluded from normal queries while remaining recoverable.	MED
SRS-DAT-006	Approval events and audit records shall be append-only.	HIGH
SRS-DAT-007	Schema changes shall be expressed as ordered, versioned migration scripts held in source control.	HIGH
SRS-DAT-008	A migration shall be additive where possible; a destructive migration shall be identified as such before application.	HIGH

Table 6.2 — Data integrity requirements.

6.3 Retention

DATA CLASS	RETENTION	BASIS
Production records	Indefinite	Project history and dispute resolution
Approval and audit events	Indefinite, append-only	Auditability (BRU-007)
Uploaded media	Indefinite unless deleted by an authorised actor	Storage capacity is the practical limit
Deactivated user records	Indefinite	Contribution history preserved (BRU-008)
Password reset tokens	30 minutes, single use	SRS-AUTH-010
Session validity cache	300 seconds maximum	SRS-SESS-007
Rate-limit counters	Duration of the window	Appendix B
Database backups	Per operator retention policy	SRS-REL-005

Table 6.3 — Retention by data class.

7 Verification

Every requirement shall be verified by at least one of the four methods below. The method is chosen by the nature of the requirement, not by convenience.

METHOD	DEFINITION	APPLIED TO
Inspection	Examination of source, configuration or documentation without executing the system.	Quality attributes, data integrity constraints, retention definitions.
Demonstration	Operation of the system to observe a stated behaviour, without measurement.	User interface requirements, workflow progression, administrative functions.
Test	Execution against defined inputs with results compared to expected values.	Authentication, authorisation, import, session and security requirements.
Analysis	Reasoning from measurement, modelling or extrapolation.	Performance and capacity requirements.

Table 7.1 — Verification methods.

7.1 Verification constraints

VER-CON-1 — VERIFICATION SHALL TRAVERSE THE REAL PATH

Any verification of a deployed behaviour shall exercise the same network path, authentication mechanism and configuration as production traffic. Verification through a privileged local channel that bypasses a control under test shall not be accepted as evidence.

Basis. A database credential rotation was previously verified by connecting from within the database container, which succeeded because local socket connections in that image use trust authentication and bypass password verification entirely. The real failure appeared only over the network, after the change was live. The constraint exists so this class of false-positive cannot recur.

VER-CON-2 — SCHEMA CHANGES SHALL BE VERIFIED AGAINST REAL DATA

Any migration or data-layer change shall be executed against a clone restored from a production backup before being applied to the live system, and the restored clone shall be confirmed non-empty before it is used as evidence.

7.2 Verification matrix by requirement class

REQUIREMENT CLASS	PRIMARY METHOD	ACCEPTANCE
Authentication (4.1)	Test	Each requirement exercised with valid, invalid and boundary inputs.
Session management (4.2)	Test	Revocation confirmed by observing a formerly-valid token rejected after each triggering event.
Access control (4.3)	Test	Every capability-gated endpoint attempted by a role lacking that capability and refused.
Tenancy (4.4)	Test	Cross-tenant access attempted for each entity type and answered as not found.
Visibility scoping (4.5)	Test	Each role's result set compared against its expected scope.
Import (4.8)	Test	Verified against real studio workbooks, including multi-sheet, flat, and malformed inputs.
Review player (4.11)	Demonstration	Each tool and comparison mode operated on a real version.
Approval chain (4.12)	Test + demonstration	Full traversal by role, with unauthorised transitions attempted and refused.
Client access (4.13)	Test	Access attempted without a code, with a revoked link, and for unshared items.
Performance (5.1)	Analysis	Measured at stated volumes on reference hardware.
Security (5.2)	Test + inspection	Each requirement individually evidenced; negative cases required.
Reliability (5.3)	Demonstration	Cache loss, container restart and restore-from-backup each exercised.

Table 7.2 — Verification approach by requirement class.

8 Appendix A — Capability reference

The fourteen capabilities and their allocation. This table is the authoritative statement of the access model; the runtime grants are held in the database and shall correspond to it.

CAPABILITY	ARTIST	LEAD	PROD. HEAD	PRODUCER	ADMIN	CLIENT	GOVERNS
create_tasks	—	Yes	Yes	Yes	—	—	Task creation, import
edit_tasks	Yes	Yes	Yes	Yes	—	—	Task field modification
assign_tasks	—	Yes	Yes	Yes	—	—	Assignee changes
delete_tasks	—	—	Yes	Yes	—	—	Task deletion
submit_reviews	Yes	Yes	Yes	Yes	—	Yes	Media upload, review submission, annotation
approve_reviews	—	Yes	Yes	Yes	—	Yes	Stage approval
broadcast_updates	—	—	Yes	Yes	—	—	Studio-wide announcements
view_financials	—	—	Yes	Yes	Yes	—	Financial views
edit_financials	—	—	Yes	Yes	—	—	Financial modification
manage_pipeline	—	—	Yes	Yes	Yes	—	Department order, stages, rollback
manage_members	—	—	Yes	Yes	Yes	—	Account lifecycle
manage_roles	—	—	Yes	Yes	Yes	—	Capability grants
manage_licenses	—	—	Yes	Yes	Yes	—	Licence administration
manage_integrations	—	—	Yes	Yes	Yes	—	Integration configuration

Table 8.1 — Capability allocation by role. The administrator column is deliberately empty of every production capability, satisfying SRS-RBAC-005 and BRU-003.

9 Appendix B — Configuration parameters

Values that govern specified behaviour. A change to any of these changes the system's conformance to the requirement that references it, and shall be treated as a specification change.

9.1 Session and credential parameters

PARAMETER	VALUE	GOVERNED BY
Session token lifetime	7 days	SRS-SESS-002
Session cookie flags	HttpOnly, SameSite=Lax	CI-002
Session validity cache TTL	300 s	SRS-SESS-007
Password hash algorithm	Argon2id	SRS-SEC-001
Password reset token lifetime	30 minutes	SRS-AUTH-010
Password reset token storage	SHA-256 hash	SRS-AUTH-009

Table 9.1 — Session and credential parameters.

9.2 Rate-limit parameters

CONTROL	KEYED BY	LIMIT	WINDOW	GOVERNED BY
Sign-in	Source address	60	900 s	SRS-SEC-004
Sign-in	Account	8	900 s	SRS-SEC-004
Registration	Source address	20	3600 s	SRS-SEC-006
Password reset request	Source address	5	900 s	SRS-SEC-006
Password reset consumption	Source address	10	900 s	SRS-SEC-006
Password change	Account	8	900 s	SRS-SEC-006

Table 9.2 — Rate-limit configuration. Address-keyed limits are sized for a whole studio behind one egress address, per SRS-SEC-007.

9.3 Upload and presence parameters

PARAMETER	VALUE	GOVERNED BY
Video upload size limit	500 MB	SRS-VER-004
Attachment size limit	200 MB	SRS-VER-004
Accepted video types	mp4, webm, mov, avi	SRS-VER-003
Presence write throttle	45 s	SRS-ATT-005
Online freshness window	90 s	SRS-ATT-006
Import parsed row ceiling	5,000 rows	SRS-IMP-007
Trusted proxy hops	1	CI-005

Table 9.3 — Upload, presence and import parameters.

10 Appendix C — Traceability matrix

Each business requirement from SYM-FRG-REQ-001 traced forward to the specification sections that satisfy it. A specification requirement with no business antecedent indicates unagreed scope; a business requirement with no specification indicates a gap.

BUSINESS REQ.	STATEMENT (ABBREVIATED)	SPECIFICATION COVERAGE
BR-001	Single authoritative record of shot status	4.6 SRS-STR-001...005; 4.7 SRS-GRD-001...007; 4.9 SRS-TSK-001...010
BR-002	Feedback bound to version and frame	4.10 SRS-VER-001, 009; 4.11 SRS-PLR-003...006, 011
BR-003	Approvals recorded with actor and time	4.12 SRS-APR-001...009; 6.2 SRS-DAT-006; BRU-007
BR-004	Existing tracksheets importable	4.8 SRS-IMP-001...012
BR-005	Media remains on studio infrastructure	4.10 SRS-VER-005...008; 5.3 SRS-REL-001, 006; BRU-006
BR-006	No per-user licensing cost	Satisfied by architecture; no licensing mechanism is specified or present
BR-007	Access restricted by role and department	4.3 SRS-RBAC-001...008; 4.4 SRS-TEN-001...004; 4.5 SRS-SCP-001...007
BR-008	External client review without an account	4.13 SRS-CLI-001...010
BR-009	Effort recorded per shot	4.14 SRS-ATT-001...011
BR-010	Runs on studio hardware without internet	2.5 CON-1, CON-2; 5.3 SRS-REL-001...007
BR-011	Oversight separable from production authority	4.3 SRS-RBAC-005; 4.9 SRS-TSK-006; 4.11 SRS-PLR-016; BRU-003; Appendix A
BR-012	Self-service onboarding, authority still granted by an administrator	4.1 SRS-AUTH-006 ... 006g, 007
BR-013	Historical changes reversible	4.15 SRS-ADM-005, 006
BR-014	Studio-wide workload visibility	4.7 SRS-GRD-001...004; 4.14 SRS-ATT-008, 010
BR-015	Third-party plug-in extension	4.15 SRS-ADM-009 — surface only; capability deferred
BR-016	DCC integration	Not specified — excluded from scope

Table 10.1 — Business requirement to specification traceability.

EXTENDING THIS MATRIX

Quality assurance shall extend Table 10.1 with a third dimension recording, for each specification requirement, the test case or verification record that evidences it. A requirement without such a record shall be treated as unverified regardless of whether the behaviour is believed to work.